
IB Mathematics HL

Paper 3 — Guided Lesson

3-Hour Interactive Session

Topics Covered

- **Hour 1** — Recurrence Relations & Sequences
- **Hour 2** — Introduction to Integration
- **Hour 3** — Hyperbolic Functions & Advanced Integration

How This Lesson Works

- ▶ Read each **Theory** box *before* attempting the question
- ★ Open the **Hint** box only when you are stuck
- Complete the question in the **work space** provided
- ✓ **Tell your teacher** when done — do not move on alone

Student name: _____

Date: _____

Teacher: _____

LESSON OVERVIEW

Hour 1 Sequences & Recurrence Relations 60 min
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- **Theory:** Arithmetic & geometric sequences, recurrence relations, S_n formulas
- **Q 1a:** Arithmetic case ($a = 1$) — 3 marks
- **Q 1b:** Geometric case ($b = 0$) — 3 marks
- **Q 1c:** Towers of Hanoi + Proof by Mathematical Induction — 9 marks
- **Q 1d/e:** General recurrence relation + verification — 10 marks

Hour 2 | Introduction to Integration 60 min

- **Theory:** Integration by substitution, standard integral $\int \frac{1}{\sqrt{1-x^2}} dx$
- **Q 2a:** Definite integral using technology — 2 marks
- **Q 2b:** Standard result for $\int \frac{1}{\sqrt{1-x^2}} dx$ — 1 mark
- **Q 2c:** Proving it via substitution $x = \sin \theta$ — 4 marks
- **Theory:** Hyperbolic functions: $\sinh x$, $\cosh x$, $\operatorname{arsinh} x$
- **Q 2d/e:** Identities and derivatives of hyperbolic functions — 6 marks

Hour 3 | Advanced Integration (4 Methods) 60 min

- **Q 2f:** Deriving $\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$ — 5 marks
- **Q 2g/h:** Method 1 — Hyperbolic substitution — 6 marks
- **Theory:** Integrating $\sec \theta$
- **Q 2i/j/k:** Method 2 — Trigonometric substitution — 6 marks
- **Q 2l/m:** Method 3 — Algebraic (no substitution) — 5 marks

HOUR 1 | Sequences & Recurrence Relations

Approximate time: 60 minutes

► THEORY — Recurrence Relations and Sequences

A **recurrence relation** defines each term of a sequence using the previous term. Today we study:

$$u_{n+1} = a u_n + b, \quad a, b \text{ constants, } u_1 \text{ given}$$

Two special cases:

Condition	Recurrence	Sequence type	Explicit formula u_n
$a = 1$	$u_{n+1} = u_n + b$	Arithmetic	$u_n = u_1 + (n - 1)b$
$b = 0, a \neq 1$	$u_{n+1} = a u_n$	Geometric	$u_n = u_1 \cdot a^{n-1}$

Sum of first n terms S_n :

- Arithmetic: $S_n = \frac{n}{2}(2u_1 + (n - 1)b)$
- Geometric ($a \neq 1$): $S_n = \frac{u_1(a^n - 1)}{a - 1}$

Quick tip: To spot the type, ask:

- Am I *adding* the same number each time? → Arithmetic
- Am I *multiplying* by the same number each time? → Geometric

QUESTION 1 — CONTEXT (Read carefully before parts a and b)

Consider a sequence with first term u_1 , defined by the recurrence relation

$$u_{n+1} = a u_n + b,$$

where a and b are constants.

For *each* of the criteria given in parts a and b:

- State the type of sequence defined by the recurrence relation.
- Write down an expression for u_n in terms of u_1 and b .
- Write down an expression for S_n (sum of first n terms) in terms of u_1 and b .

QUESTION 1a

(3 marks)

$$a = 1$$

★ HINT — Q1a — the arithmetic case

If $a = 1$, the relation becomes $u_{n+1} = u_n + b$.

This means each term is the previous one **plus** b — a constant addition. What do we call that?

For u_n : starting from u_1 , we add b exactly $(n - 1)$ times.

For S_n : use the arithmetic sum formula from the theory box.

i. Type of sequence: _____

ii. $u_n =$ _____

iii. $S_n =$ _____

QUESTION 1b

(3 marks)

$$b = 0, \quad a \neq 1, \quad a \neq 0$$

★ HINT — Q1b — the geometric case

If $b = 0$, the relation becomes $u_{n+1} = a u_n$.

This means each term is the previous one **multiplied** by a . What type is that?

For u_n : starting from u_1 , we multiply by a exactly $(n - 1)$ times.

For S_n : use the geometric sum formula. Remember the condition $a \neq 1$.

i. Type of sequence: _____

ii. $u_n =$ _____

iii. $S_n =$ _____

► THEORY — Proof by Mathematical Induction

Mathematical induction proves a statement $P(n)$ is true for *all* integers $n \geq 1$. It has **four essential parts**:

Step 1. Base case. Show $P(1)$ is true by direct substitution.

Step 2. Inductive hypothesis. Assume $P(k)$ is true for some arbitrary $k \geq 1$.

Step 3. Inductive step. Using *only* the assumption in Step 2, *prove* that $P(k+1)$ must be true.

Step 4. Conclusion. State: “By the principle of mathematical induction, $P(n)$ is true for all $n \geq 1$.”

Key trick for recurrence proofs:

If you want to prove $u_{k+1} = 2^{k+1} - 1$, *start* from the recurrence relation:

$$u_{k+1} = 2u_k + 1,$$

then substitute $u_k = 2^k - 1$ (your hypothesis) and simplify.

Common mistakes to avoid:

- Do **not** assume $P(k+1)$ and work backwards.
- Always start the inductive step from u_{k+1} (or the LHS of $P(k+1)$).
- You **must** write a formal conclusion sentence to earn full marks.

QUESTION 1c

(9 marks)

The *Towers of Hanoi* is a famous example of a sequence defined by a recurrence relation of the form $u_{n+1} = au_n + b$.

In this sequence, $u_1 = 1$ and $u_{n+1} = 2u_n + 1$.

- Find u_2 , u_3 , u_4 and u_5 .
- Using your answers to part **i**, suggest an explicit expression for u_n in terms of n .
- Prove your expression from part **ii** using mathematical induction.

★ HINT — Q1c

Part i: Apply $u_{n+1} = 2u_n + 1$ starting with $u_1 = 1$:
 $u_2 = 2(1) + 1 = 3$, $u_3 = 2(3) + 1 = ?$, keep going...

Part ii: List your terms: 1, 3, 7, 15, 31, ...
 Write each as a power of 2 minus something. Can you see it?

Part iii — Induction outline:

$P(n)$: “ $u_n = 2^n - 1$ ”

1. *Base:* $u_1 = 2^1 - 1 = 1$ ✓

2. Assume: $u_k = 2^k - 1$ for some $k \geq 1$.

3. Step: $u_{k+1} = 2u_k + 1 = 2(2^k - 1) + 1 = 2^{k+1} - 2 + 1 = 2^{k+1} - 1 \checkmark$

4. Conclude.

Part i:

n	1	2	3	4	5
u_n	1				

Part ii: $u_n =$ _____

Part iii — Proof by Mathematical Induction:

Let $P(n)$ be the statement: $u_n =$ _____

Base case ($n = 1$):

Inductive hypothesis:

Assume $P(k)$ is true for some $k \geq 1$, i.e. $u_k =$ _____

Inductive step (prove $P(k + 1)$ is true):

Conclusion:

QUESTION 1d

(8 marks)

When $a \neq 1$, the recurrence relation $u_{n+1} = au_n + b$ can be rewritten in the form

$$u_{n+1} + c = a(u_n + c)$$

for some constant c .

- i. Find the constant c in terms of a and b .

A new sequence is defined by $v_n = u_n + c$.

- ii. Write down the recurrence relation for v_n and state what type of sequence v_n is.
 iii. Hence, write down an explicit formula for v_n in terms of v_1 , a and n .
 iv. Hence, write down an explicit formula for u_n in terms of u_1 , a , b and n .

★ HINT — Q1d

Part i: Expand the right-hand side: $a(u_n + c) = au_n + ac$.

The left-hand side: $u_{n+1} + c = (au_n + b) + c$.

For these to be equal for all n : match coefficients of the constant terms: $b + c = ac$, so $c(a - 1) = b$, so $c = \frac{b}{a - 1}$.

Part ii: By definition $v_{n+1} = u_{n+1} + c$. Use the rewritten relation to show $v_{n+1} = av_n$. This is a geometric sequence with common ratio a .

Part iii: Geometric sequence formula: $v_n = v_1 \cdot a^{n-1}$.

Part iv: Since $v_n = u_n + c$, we get $u_n = v_n - c = v_1 \cdot a^{n-1} - c$.
 Substitute $v_1 = u_1 + c$ and $c = \frac{b}{a - 1}$, then simplify.

Part i — finding c :

$c =$ _____

Part ii:

Recurrence for v_n : _____ Type: _____

Part iii:

$$v_n = \underline{\hspace{10cm}}$$

Part iv — full working:

$$u_n = \underline{\hspace{10cm}}$$

QUESTION 1e

(2 marks)

Verify that the expression you obtained for u_n in part **d(iv)** for the Towers of Hanoi is consistent with the formula you derived for u_n in part **c(ii)**.

★ **HINT — Q1e**

For the Towers of Hanoi: $a = 2$, $b = 1$, $u_1 = 1$.

Substitute these specific values into your general formula from Q 1d(iv) and show that you obtain $u_n = 2^n - 1$.

END OF QUESTION 1

[Total: 25 marks]

STOP — show your teacher your work before turning the page.

HOUR 2 | Introduction to Integration

Approximate time: 60 minutes

► THEORY — Integration by Substitution

Integration by substitution reverses the chain rule. The idea:

$$\int f(g(x)) g'(x) dx = \int f(u) du \quad \text{where } u = g(x), du = g'(x) dx.$$

Steps:

1. Choose a substitution $u = g(x)$.
2. Compute $du = g'(x) dx$ and replace dx .
3. Rewrite *everything* in terms of u .
4. Integrate, then substitute back.

Standard result to know:

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

You will prove this yourself in Q 2c!

Our goal for Question 2: Find $I = \int \frac{1}{\sqrt{1+x^2}} dx$ using **four different methods**.

#	Method	Parts
1	Technology (numerical check)	2a
2	Hyperbolic substitution $x = \sinh u$	2g, 2h
3	Trigonometric substitution $x = \tan \theta$	2i, 2j, 2k
4	Algebraic — no substitution needed	2l, 2m

QUESTION 2 — CONTEXT

In this question you will find the value of the indefinite integral

$$I = \int \frac{1}{\sqrt{1+x^2}} dx$$

using different methods. Suppose that first you are asked to find the value of a *definite* integral. You can do this using technology.

QUESTION 2a

(2 marks)

Write down the value of $\int_0^1 \frac{1}{\sqrt{1+x^2}} dx$.

★ HINT — Q2a

Use your GDC. Go to the numerical integration menu and enter the function $\frac{1}{\sqrt{1+x^2}}$ with lower limit 0 and upper limit 1.

Give your answer correct to 3 significant figures. Keep it — you will check it later using an exact answer.

$$\int_0^1 \frac{1}{\sqrt{1+x^2}} dx = \underline{\hspace{2cm}}$$

QUESTION 2b

(1 mark)

Write down $\int \frac{1}{\sqrt{1-x^2}} dx$.

★ HINT — Q2b

This is a **standard result** from your formula booklet. The function $\frac{1}{\sqrt{1-x^2}}$ is the derivative of a familiar inverse trig function.

$$\int \frac{1}{\sqrt{1-x^2}} dx = \underline{\hspace{2cm}}$$

QUESTION 2c

(4 marks)

Show how the answer in (b) can be proved by applying the substitution $x = \sin \theta$ $\left(-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}\right)$ to $\int \frac{1}{\sqrt{1-x^2}} dx$.

★ HINT — Q2c

Step by step:

1. Let $x = \sin \theta$, so $\frac{dx}{d\theta} = \cos \theta$, i.e. $dx = \cos \theta d\theta$.
2. Find $1 - x^2 = 1 - \sin^2 \theta = \cos^2 \theta$ (Pythagorean identity).
3. So $\sqrt{1 - x^2} = \cos \theta$ (positive because $-\pi/2 \leq \theta \leq \pi/2$).
4. Substitute into the integral: $\int \frac{1}{\cos \theta} \cdot \cos \theta d\theta = \int 1 d\theta = \theta + C$.
5. Convert back: $\theta = \arcsin x$, so $I = \arcsin x + C$.

Checkpoint: Have you finished Q 2c? Tell your teacher. Then read the theory box on the next page before continuing.

► THEORY — Hyperbolic Functions — sinh, cosh and arsinh

Hyperbolic functions are defined using exponentials. They have similar names and identities to trig, but are *not* the same.

Function	Definition	Derivative
$\sinh x$	$\frac{e^x - e^{-x}}{2}$	$\cosh x$
$\cosh x$	$\frac{e^x + e^{-x}}{2}$	$\sinh x$

Key identity: $\cosh^2 x - \sinh^2 x = 1$ (compare: $\cos^2 \theta + \sin^2 \theta = 1$)

Why is this useful for I?

If we let $x = \sinh u$, then $1 + x^2 = 1 + \sinh^2 u = \cosh^2 u$, so $\sqrt{1 + x^2} = \cosh u$.

The square root disappears — the integral simplifies beautifully!

Inverse function: $y = \operatorname{arsinh} x \iff x = \sinh y$

You will prove in Q 2f that $\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$.

Comparison table:

Trigonometric	Hyperbolic
$\cos^2 \theta + \sin^2 \theta = 1$	$\cosh^2 x - \sinh^2 x = 1$
$\frac{d}{d\theta}(\sin \theta) = \cos \theta$	$\frac{d}{dx}(\sinh x) = \cosh x$
$\frac{d}{d\theta}(\cos \theta) = -\sin \theta$	$\frac{d}{dx}(\cosh x) = \sinh x$ (no minus!)
$\sin^{-1} x = \arcsin x$	$\sinh^{-1} x = \operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$

QUESTION 2d

(2 marks)

Show that $(\cosh x)^2 - (\sinh x)^2 = 1$.

★ HINT — Q2d

Substitute the definitions directly:

$$\cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2$$

Expand each square. The e^{2x} and e^{-2x} terms will cancel, and the cross terms add up nicely.

QUESTION 2e
(4 marks)

Find the derivative of:

i. $\sinh x$

ii. $\cosh x$

Give your answers in terms of $\sinh x$ and $\cosh x$.

★ HINT — Q2e

Differentiate each definition term by term:

$$\frac{d}{dx} \left(\frac{e^x - e^{-x}}{2} \right) = \frac{e^x + e^{-x}}{2} = \cosh x$$

Do the same for $\cosh x$. Note: the derivative of $-e^{-x}$ is $+e^{-x}$ (chain rule with -1).

Part i: $\frac{d}{dx}(\sinh x) =$ _____

Part ii: $\frac{d}{dx}(\cosh x) =$ _____

END OF HOUR 2 MATERIAL
STOP — check with your teacher before continuing to Hour 3.

HOUR 3 | Advanced In- tegration — 4 Methods for I

Approximate time: 60 minutes

QUESTION 2f

(5 marks)

By solving a quadratic in e^x , show and justify that

$$\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1}).$$

★ HINT — Q2f

Let $y = \operatorname{arsinh} x$, so by definition $x = \sinh y = \frac{e^y - e^{-y}}{2}$.

1. Write: $2x = e^y - e^{-y}$.
2. Multiply both sides by e^y : $2xe^y = e^{2y} - 1$.
3. Rearrange to a quadratic: $e^{2y} - 2xe^y - 1 = 0$.
4. Let $t = e^y$ (to make it look like $at^2 + bt + c = 0$).
5. Quadratic formula: $t = \frac{2x \pm \sqrt{4x^2 + 4}}{2} = x \pm \sqrt{x^2 + 1}$.
6. Since $t = e^y > 0$ always, take the **positive** root: $e^y = x + \sqrt{x^2 + 1}$.
7. Take natural log of both sides: $y = \ln(x + \sqrt{x^2 + 1})$.

Justify why $x - \sqrt{x^2 + 1}$ is always negative (for all real x).

QUESTION 2g

(4 marks)

 Find the indefinite integral I by applying the substitution $x = \sinh u$.

★ HINT — Q2g

1. $x = \sinh u \Rightarrow \frac{dx}{du} = \cosh u \Rightarrow dx = \cosh u \, du$.
2. $1 + x^2 = 1 + \sinh^2 u = \cosh^2 u$ (from Q2d).
3. $\sqrt{1 + x^2} = \cosh u$ (cosh is always positive, so no absolute value needed).
4. Integral becomes: $\int \frac{1}{\cosh u} \cdot \cosh u \, du = \int 1 \, du = u + C$.
5. Convert back: $u = \operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$ (from Q2f).

 $I =$ _____

QUESTION 2h

(2 marks)

 Hence, find the exact value of $\int_0^1 \frac{1}{\sqrt{1+x^2}} dx$ and verify that it agrees with the answer from part (a).

★ HINT — Q2h

Apply the limits to your result from Q2g:

$$\left[\ln(x + \sqrt{x^2 + 1}) \right]_0^1 = \ln(1 + \sqrt{2}) - \ln(0 + \sqrt{1}) = \ln(1 + \sqrt{2}) - \ln 1 = \ln(1 + \sqrt{2})$$

 Evaluate $\ln(1 + \sqrt{2})$ on your GDC and compare with Q2a.

Exact value: _____ Decimal (3 s.f.): _____ Consistent with (a)? _____

► THEORY — Integrating $\sec \theta$ — a clever trick

To evaluate $\int \sec \theta \, d\theta$, multiply by a clever form of 1:

$$\int \sec \theta \, d\theta = \int \sec \theta \cdot \frac{\sec \theta + \tan \theta}{\sec \theta + \tan \theta} \, d\theta = \int \frac{\sec^2 \theta + \sec \theta \tan \theta}{\sec \theta + \tan \theta} \, d\theta$$

Now notice: the **numerator is exactly the derivative of the denominator**.

Let $t = \sec \theta + \tan \theta$, then $\frac{dt}{d\theta} = \sec \theta \tan \theta + \sec^2 \theta$.

$$\int \frac{1}{t} \, dt = \ln |t| + C = \ln |\sec \theta + \tan \theta| + C$$

Result: $\int \sec \theta \, d\theta = \ln |\sec \theta + \tan \theta| + C$

You will derive this step by step in Q 2j.

QUESTION 2i

(2 marks)

Apply the substitution $x = \tan \theta$ $\left(-\frac{\pi}{2} < \theta < \frac{\pi}{2}\right)$ to show that I can be changed to $\int \sec \theta \, d\theta$.

★ HINT — Q2i

1. $x = \tan \theta \Rightarrow dx = \sec^2 \theta \, d\theta$.
2. $1 + x^2 = 1 + \tan^2 \theta = \sec^2 \theta$ (standard trig identity).
3. $\sqrt{1 + x^2} = \sec \theta$ (positive for $-\pi/2 < \theta < \pi/2$).
4. Substitute: $I = \int \frac{1}{\sec \theta} \cdot \sec^2 \theta \, d\theta = \int \sec \theta \, d\theta$.

QUESTION 2j

(2 marks)

Multiply $\sec \theta$ by 1, written as $\frac{\sec \theta + \tan \theta}{\sec \theta + \tan \theta}$, to obtain the answer to $\int \sec \theta d\theta$ in terms of θ .

★ HINT — Q2j

Follow the steps in the Theory box above. Show that the numerator of the resulting fraction is the derivative of the denominator, then use the standard result $\int \frac{f'}{f} dx = \ln |f| + C$.

$$\int \sec \theta d\theta = \underline{\hspace{10em}}$$

QUESTION 2k

(2 marks)

Express the answer found in part **j** in terms of x , and verify this is consistent with the expression you obtained for I in part **g**.

★ HINT — Q2k

From $x = \tan \theta$, draw a right triangle or use identities to find $\sec \theta$ in terms of x :
 $\sec \theta = \sqrt{1 + \tan^2 \theta} = \sqrt{1 + x^2}$ and $\tan \theta = x$.
 Substitute: $I = \ln |\sec \theta + \tan \theta| + C = \ln(\sqrt{1 + x^2} + x) + C$.
 Compare with Q 2g: $I = \ln(x + \sqrt{x^2 + 1}) + C$. They should match!

QUESTION 2l

(2 marks)

Let $f(x) = x + \sqrt{1 + x^2}$. Find $f'(x)$.

★ HINT — Q2l

Differentiate term by term using the chain rule on $\sqrt{1+x^2} = (1+x^2)^{1/2}$:

$$\frac{d}{dx}(\sqrt{1+x^2}) = \frac{1}{2}(1+x^2)^{-1/2} \cdot 2x = \frac{x}{\sqrt{1+x^2}}$$

So $f'(x) = 1 + \frac{x}{\sqrt{1+x^2}}$.

Combine over a common denominator to get a single fraction.

$$f'(x) = \underline{\hspace{10cm}}$$

QUESTION 2m

(3 marks)

Multiply $\frac{1}{\sqrt{1+x^2}}$ by 1, written as $\frac{f'(x)}{f(x)}$, using your expression for $f'(x)$ from part 1. Hence simplify the expression for I into an expression involving only $f(x)$ and $f'(x)$. In this way, find the integral I directly.

★ HINT — Q2m

Key insight: If you can show $\frac{f'(x)}{f(x)} = \frac{1}{\sqrt{1+x^2}}$, then:

$$I = \int \frac{1}{\sqrt{1+x^2}} dx = \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C = \ln(x + \sqrt{1+x^2}) + C$$

To verify: Show that $\frac{f'(x)}{f(x)} = \frac{1}{\sqrt{1+x^2}}$ by substituting your $f(x)$ and $f'(x)$ and simplifying.

Hint: when simplifying $\frac{f'(x)}{f(x)}$, try factoring $\frac{1}{\sqrt{1+x^2}}$ from the numerator.

$I =$ _____

END OF QUESTION 2 [Total: 35 marks]

Section	Your score	Out of	Percentage
Question 1	_____	25	_____
Question 2	_____	35	_____
TOTAL	_____	60	_____

KEY RESULTS SUMMARY

For revision and reference

✓ SUMMARY — Recurrence Relations — $u_{n+1} = au_n + b$

Condition	Type	u_n	S_n
$a = 1$	Arithmetic	$u_1 + (n - 1)b$	$\frac{n}{2}(2u_1 + (n - 1)b)$
$b = 0, a \neq 1$	Geometric	$u_1 \cdot a^{n-1}$	$\frac{u_1(a^n - 1)}{a - 1}$
General ($a \neq 1$)	—	$\left(u_1 + \frac{b}{a - 1}\right)a^{n-1} - \frac{b}{a - 1}$	—

Towers of Hanoi: $a = 2, b = 1, u_1 = 1 \Rightarrow u_n = 2^n - 1$ (proved by induction)

✓ SUMMARY — Hyperbolic Functions

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

$$\cosh^2 x - \sinh^2 x = 1, \quad \frac{d}{dx}(\sinh x) = \cosh x, \quad \frac{d}{dx}(\cosh x) = \sinh x$$

$$\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$$

✓ SUMMARY — The Integral $I = \int \frac{1}{\sqrt{1 + x^2}} dx$

All four methods give the same answer:

$$I = \ln(x + \sqrt{x^2 + 1}) + C = \operatorname{arsinh} x + C$$

Definite integral:

$$\int_0^1 \frac{1}{\sqrt{1 + x^2}} dx = \ln(1 + \sqrt{2}) \approx 0.881$$

Method	Substitution / Trick
Hyperbolic	$x = \sinh u$, use $1 + \sinh^2 u = \cosh^2 u$
Trig	$x = \tan \theta$, use $1 + \tan^2 \theta = \sec^2 \theta$, then integrate $\sec \theta$
Algebraic	Write integrand as $\frac{f'(x)}{f(x)}$ where $f(x) = x + \sqrt{1 + x^2}$

Well done — you have completed the full Paper 3 lesson!

Review any questions you found difficult before your next practice paper.